WAYAMBA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

B.Sc. (GENERAL/JOINT MAJOR/ SPECIAL) DEGREE EXAMINATION
Level 3 Semester I– (Academic Year 2021/2022) Jan/Feb 2024

Department of Computing and Information Systems

CMIS 3142 – Computational Methods

No. of Pages: Three (03) *No. of Questions: Six (06)* *Time: Two (02) hours*

Answer four(4) Questions only

Calculators are not permitted

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1)

a) Summarize the range of numbers that can be represented by the IEEE 754 single precision and double precision.

b) Describe the difference between normalized and denormalized numbers.

c) Write a Java program to convert a given decimal number into the IEEE 754 double precision format and vice versa.

d) Write a Java program to convert the IEEE 754 single precision format to IEEE 754 double precision format and vice versa. Mention the assumptions, if any.

2)

a) Describe the relative error and true fractional relative error.

b) Describe the maximum and minimum rounding off error, with example, in the calculation of the final GPA (Mention the assumptions, if any). Propose a method to minimize the above error.

c) Find

i) Δe^{ax} ii) $\Delta^2 e^x$ iii) $\Delta \log(x)$

d) Evaluate $\Delta^2 \frac{1}{x}$ by taking '1' as the interval of differencing.

3)

Using inheritance and polymorphism, write one Java class for each of the following matrix categories with the matrix operations given below:

i) Full matrix
ii) Symmetric matrix
iii) Sparse matrix
iv) Sparse symmetric matrix
v) Row vector
vi) Column vector

Matrix operations to be considered for the above are given below:

i. Addition
ii. Subtraction
iii. Multiplication
iv. Row swap
v. Column swap

4)

a) Write a numerical algorithm for Jacobi method

b) Write a numerical algorithm for Gauss-Seidel method.

c) Solve the following equations using Jacobi method

i) $x + y + z = 3$
ii) $2x - y - z = 3$
iii) $x - y + z = 9$

d) Solve the above equations using the Gauss-Seidel method.
e) Compare and contrast Jacobi and Gauss-Seidel method

5)

a) Develop an algorithm to solve a given system of linear equation.

b) Solve the following equations using SOR (Successive over-relaxation) method (with initial guess (0,0,0) and w = 1.25)

i) $10x + 2y - z = 7$
ii) $x + 8y + 3z = -4$,
iii) $-2x - y + 10z = 9$

c) Solve the following equations using the Cholesky decomposition method

i) $25x + 15y - 5z = 35$,
ii) $15x + 18y = 33$
iii) $-5x + 11z = 6$

6)

a) Describe the Gaussian elimination method to solve given system of liner equations.

b) There is a contractor who has a set of workers for his job and he defines the age group of people as follows:

No	Group	Gender	Age range
1	Teen	Male and female	13 – 19
2	Man	Male	20 – 55
3	Woman	Female	20 – 50
4	Old	Male	Greater than 55

He gave his quotation to complete a job as follow:

8 teens, 2 men, 3 women, and 2 old will cost 189 $ per day.
3 teens, 6 men, 2 women, and 1 old will cost 192 $ per day.
1 teen, 2 men, 9 women, and 4 old will cost 233 $ per day.
2 teens, 3 men, 4 women, and 7 old will cost 224 $ per day.

Payment for each category per day was not mentioned in the quotation.

i) Convert the above real world problem into system of linear equations to find the payment of each category per day.

ii) Solve the problem using the Gaussian elimination method.